Report for Silicon Photonics Design, Fabrication and Data Analysis edX course

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Primary goal 1	Extract the waveguide group index n_g from the spectrum of the unbalanced interferometer and compare the result with the waveguide simulation baseline and process-corner bounds.
Primary goal 2	Extract passive-component losses and spectral agreement metrics for crossings, bends, splitters, and couplers using normalized experiment-versus-simulation comparisons.
Supporting goal	Use a single-bus ring resonator as an additional dispersion-sensitive structure.

1. Introduction

Silicon photonics provides a compact platform for waveguides, splitters, couplers, interferometers, and resonators fabricated on silicon-on-insulator wafers. In this chip, the central characterization questions are dispersion and loss: the dispersion question is addressed by extracting the group index from an unbalanced Mach-Zehnder interferometer and checking it against a ring resonator and simulated waveguide data, while the loss question is addressed through dedicated crossing, bend, splitter, and coupler test structures.

2. Chip layout and structure inventory

Figure 1 shows the full chip layout used in the analysis package. The chip combines the primary unbalanced MZI, a ring resonator, a reference waveguide, and a set of passive benchmark structures that isolate crossings, bends, Y-splitters, a directional coupler, a pi-phase-shift MZI, an MMI 1x4, and a two-stage Y-splitter cascade. The wide structure coverage is important because group-index extraction and passive-loss extraction depend on the same compact-model and normalization assumptions.

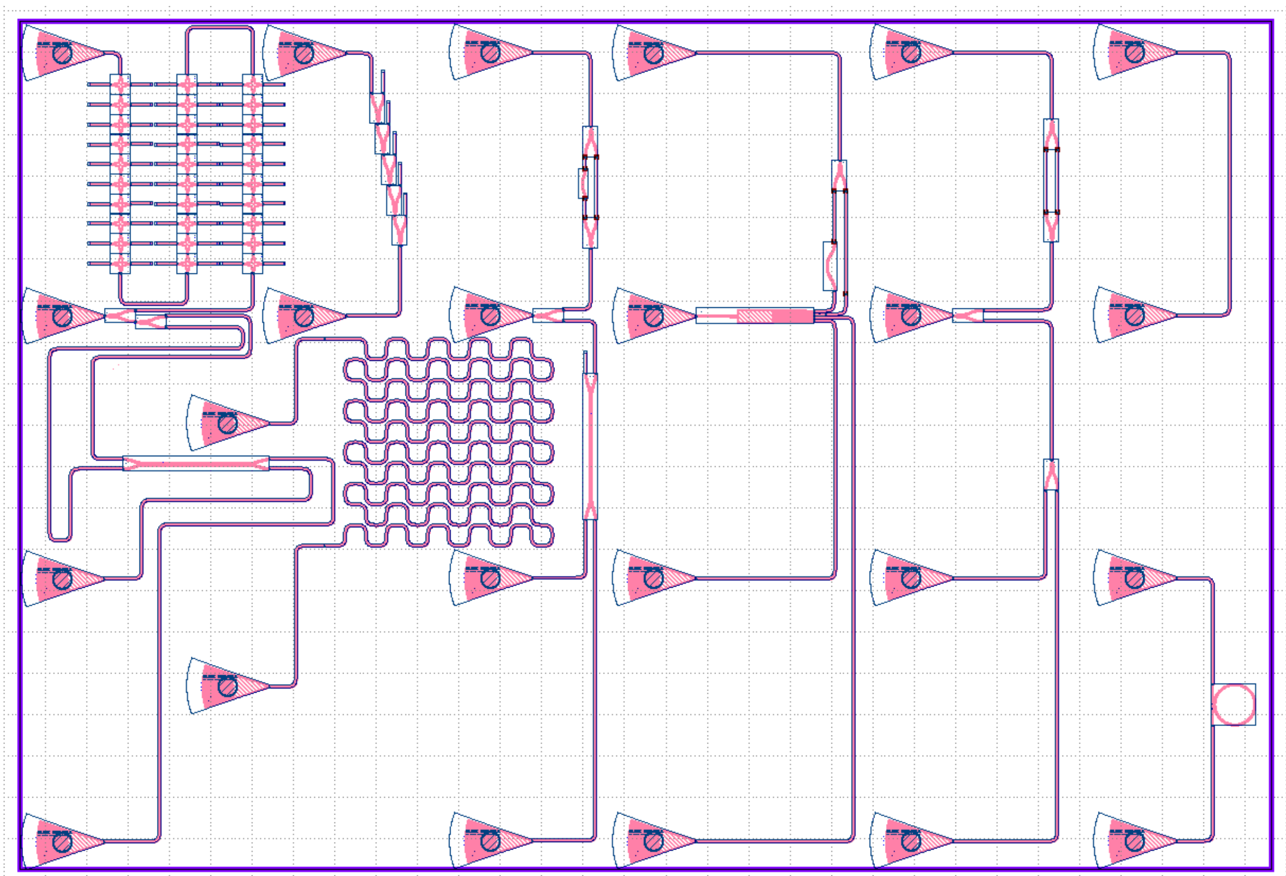


Figure 1. Full chip layout used for the experimental analysis. The layout contains the unbalanced MZI, ring resonator, reference structure, and dedicated passive benchmark devices.

Structure	Purpose	Main observable
Reference waveguide	Define the grating-coupler envelope used for normalization	Broad transmission envelope versus wavelength
Unbalanced MZI (Delta L = 100 um)	Primary group-index extraction structure	FSR, paired-port analytic fit, local n_g checkpoints
Single-bus ring resonator	Independent resonant cross-check of dispersion	Resonance spacing / ring FSR

30 crossings	Crossing-loss extraction	Insertion loss per crossing
200 bends	Bend-loss extraction	Insertion loss per bend
Five-stage Y-splitter tree	Excess-loss extraction for Y-splitters	Transmission of a 1-to-32 splitter tree
Directional coupler	De-embed the coupler used in the MZI	Two-output spectral balance
Pi-phase-shift MZI	Check phase-targeted destructive interference	Null depth near 1550 nm
MMI 1x4	Check output balance and destructive top-port recombination	Bottom-output balance and top-port suppression
Two-stage Y-splitter cascade	Reference comparison against the MMI 1x4	Top-output transmission and bottom-output balance

3. Modeling and data-processing framework

3.1 Waveguide compact model from the uploaded simulation text files

The real part of n_eff(lambda) is fit with a quadratic compact model around lambda0 = 1.55 um: n_eff(lambda) = n1 + n2(lambda-lambda0) + n3(lambda-lambda0)^2. The corresponding group index is evaluated as n_g(lambda) = n_eff(lambda) - lambda d n_eff / d lambda.

At 1550 nm the updated simulation values are n_eff = 2.444578 and n_g = 4.2019, and the fitted coefficients are n1 = 2.444579, n2 = -1.133756 1/um, and n3 = -0.062360 1/um^2. The nominal slope d n_g / d lambda at 1550 nm is 1.93 x 10^-4 per nm.

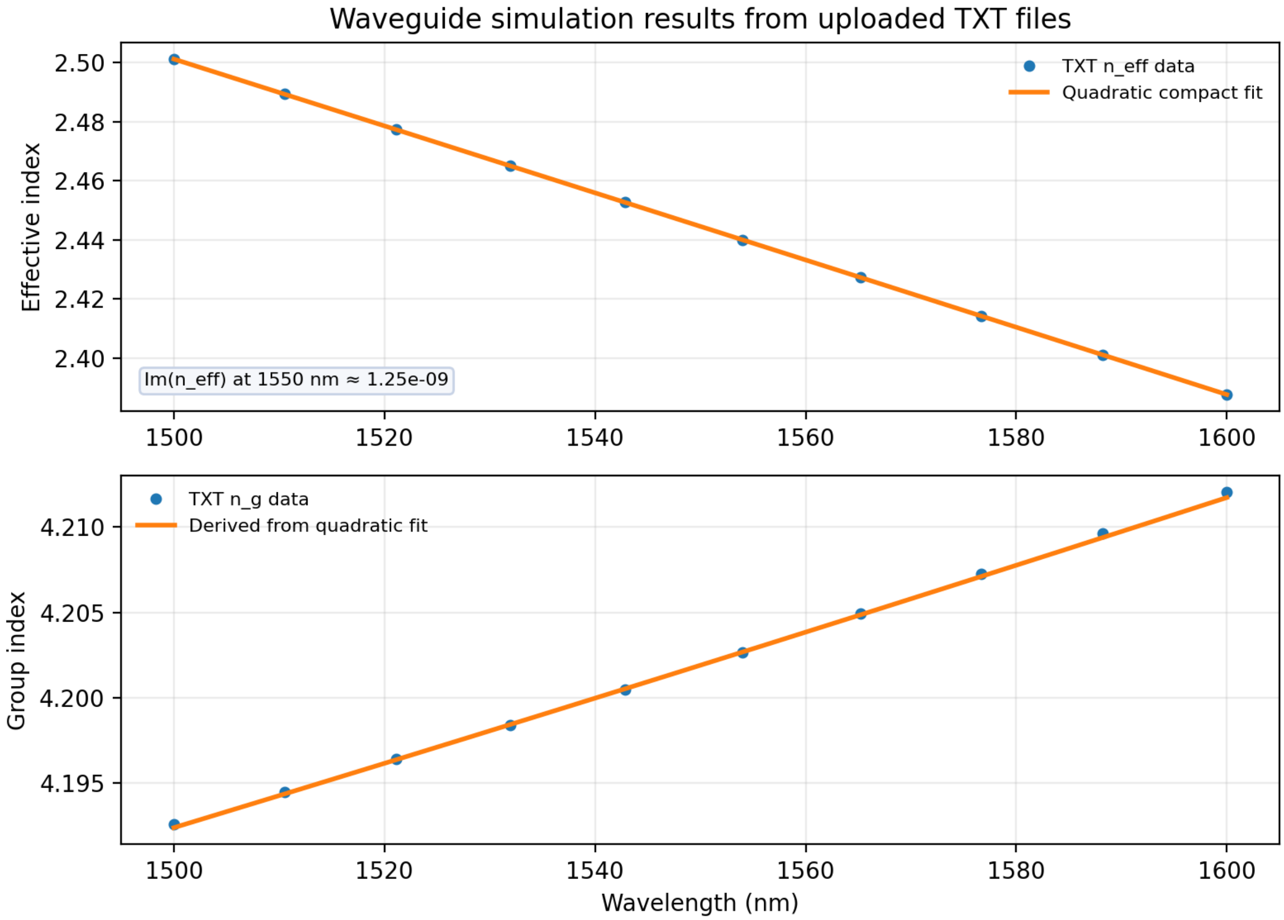


Figure 2. Nominal waveguide baseline reconstructed from the uploaded text files: real and imaginary n_eff(lambda), uploaded n_g(lambda), and the quadratic compact fit used as the nominal dispersion baseline.

Metric	Value	Metric	Value
n_eff(1500 nm)	2.50111	n_g(1500 nm)	4.1926
n_eff(1550 nm)	2.444578	n_g(1550 nm)	4.2019
Im(n_eff)(1550 nm)	1.249e-09	n_g(1600 nm)	4.2120
Quadratic n1	2.444579	Quadratic n2 [1/um]	-1.133756
Quadratic n3 [1/um^2]	-0.062360	d n_g / d lambda at 1550 [1/nm]	0.000193

3.2 MZI and ring relations used for group-index extraction

For the unbalanced interferometer, the spectral period is related to group index through FSR_MZI approximately equal to lambda^2 / (n_g Delta L). In the present chip, Delta L = 100 um is the relevant MZI imbalance for the final report, which gives a nominal simulated FSR of 5.72 nm at 1550 nm using the corrected nominal waveguide baseline. For the ring resonator, FSR_ring approximately equals lambda^2 / (n_g 2 pi R), and with R = 10 um the corrected nominal FSR at 1550 nm is approximately 9.10 nm.

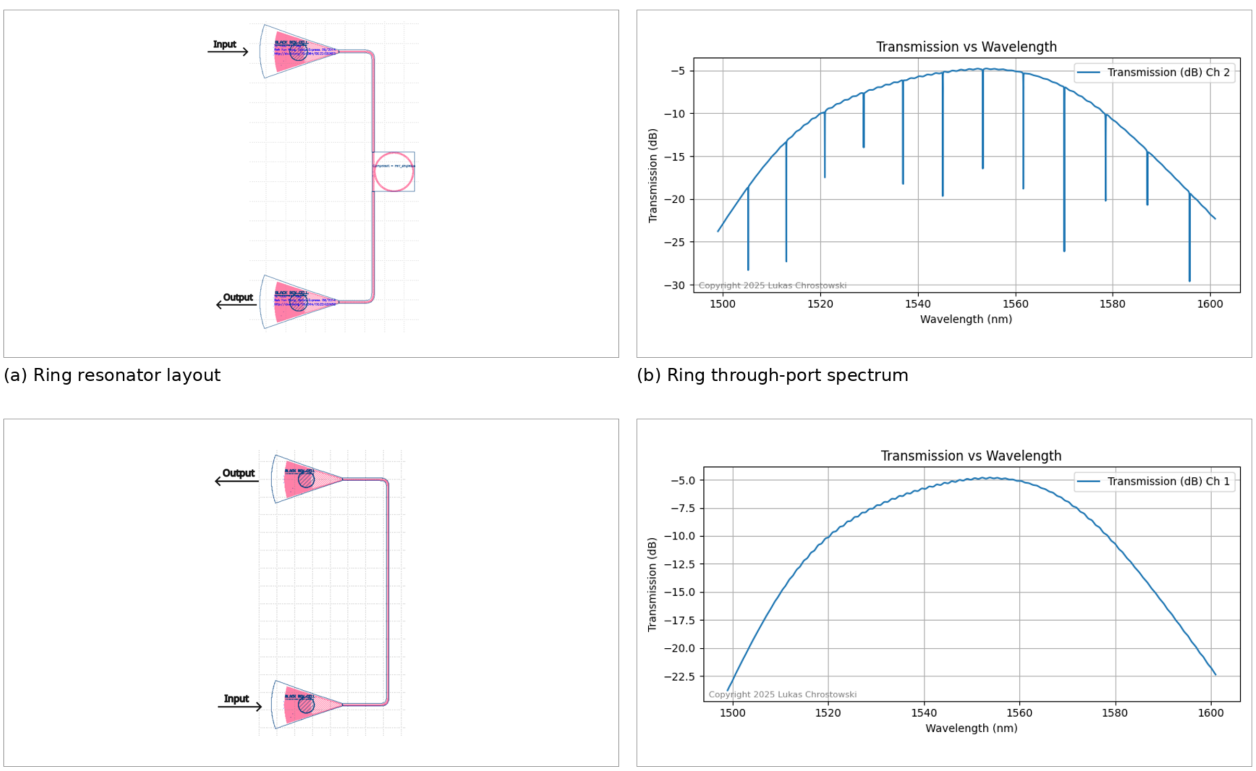
The MZI fitting script uses the complementary outputs jointly. A shared polynomial n_eff model is fit to the two ports with a sign flip between the ports, and a local find-peaks routine separately estimates FSR and pointwise n_g from the minima spacing. The smooth paired-port fit is therefore treated as the primary global estimate, while the pointwise minima-based values are treated as local checkpoints. Process-corner n_g curves are overlaid to show whether the measured result is contained within the simulated corner band.

3.3 Reference-envelope normalization and simulation alignment

Measured device spectra are not compared against simulation directly in their raw form. Instead, the reference waveguide - two grating couplers connected by a waveguide - is used to estimate the grating-coupler envelope. The normalization pipeline fits a fourth-order polynomial to the top 10 dB of the reference passband and subtracts that fitted envelope from each measured device spectrum in dB. In compact form, T_norm,dB(lambda) = T_device,dB(lambda) - T_ref,fit,dB(lambda). For the MZI and pi-phase-shift MZI branches, the code also removes the nominal 3 dB Y-splitter factor so that the normalized traces represent the device response rather than the splitter tree feeding it.

Simulation traces are interpolated onto the same wavelength grid as the normalized measured traces. This makes experiment-versus-simulation comparisons meaningful on both the linear-power and dB scales and enables direct calculation of mean absolute error (MAE) in dB and mean linear-power deviation in percent. The same codebase supports both Scylla and Iris datasets through dataset-specific channel maps, normalization plans, and simulation-channel assignments.

Ring-resonator and reference structures



(a) Ring resonator layout

(b) Ring through-port spectrum

(c) Reference structure layout

(d) Reference transmission envelope

Figure 3. Reference-envelope normalization context. The reference structure defines the normalization envelope, while the ring resonator provides an additional dispersion-sensitive spectrum that can be checked against the nominal ring FSR.

4. Results

4.1 Unbalanced MZI group-index extraction

The MZI result is the centerpiece of the report. The complementary-port analytic fits reproduce the measured fringe pattern well, the local FSR values stay in the expected 5.7-5.9 nm band, and the final group-index comparison overlays the paired-port fit, the minima-based checkpoints, the nominal simulation, and the process-corner bounds in one view. This is the essential figure for the group-index objective because it shows not only the extracted value but also how that value relates to the nominal and corner simulation baselines.

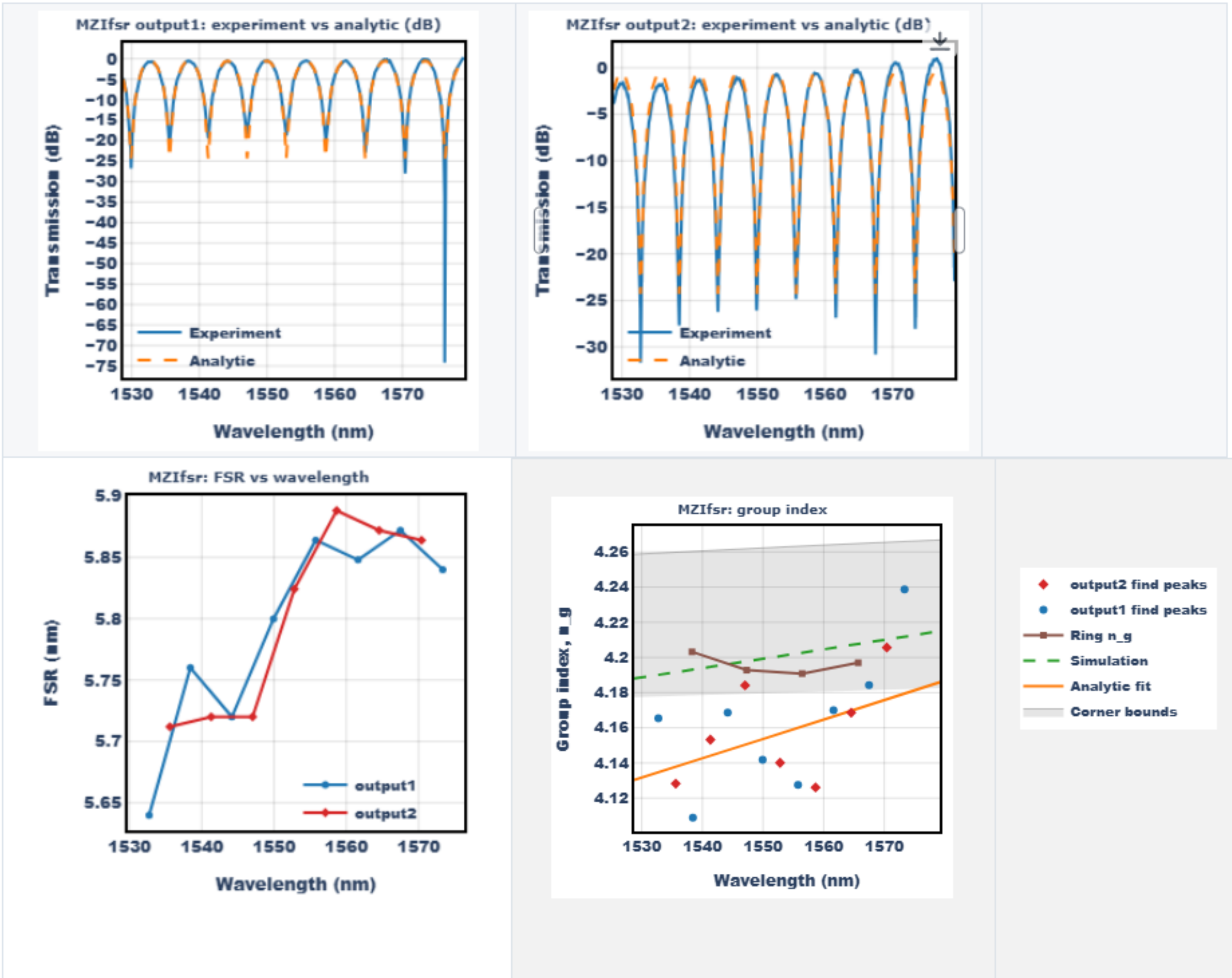


Figure 4. Essential MZI summary figure: complementary-port analytic fits, FSR from minima spacing, and overlay of minima-based n_g, paired-port analytic-fit n_g, nominal simulation, and simulated process-corner bounds.

Estimate	n_g(1550 nm)	n_g(lambda) range	Interpretation
Find-peaks output 1	4.1417	[4.1090, 4.2387]	1.43% below nominal simulation
Find-peaks output 2	4.1615	[4.1261, 4.2057]	0.96% below nominal simulation
Analytic paired-port fit	4.1536	[4.1305, 4.1859]	Primary global estimate; 1.15% below nominal simulation
Nominal simulation	4.2019	[4.1926, 4.2120]	Corrected baseline from uploaded n_g(lambda) text file
Lower corner bound	4.1797	[4.1774, 4.1826]	Simulated lower process corner

Upper corner bound	4.2623	[4.2587, 4.2668]	Simulated upper process corner

Three facts matter most. First, the two minima-based outputs and the analytic paired-port fit are mutually consistent: all three measured estimates lie in the narrow 4.14-4.16 range at 1550 nm. Second, all measured estimates sit below the corrected nominal simulation and even below the lower simulated corner bound. This means the remaining mismatch is not a plotting artifact; it reflects a real gap between the circuit-level prediction and the extracted device response. Third, the broken minima-based n_g points do not contradict the smooth analytic trend. They are the expected high-variance output of a local estimator that uses only a handful of minima spacings, whereas the analytic fit uses the full trace shape.

The most likely explanations for the residual MZI gap are a small effective Delta L error, unmodeled directional-coupler phase contribution, and residual bias in the reference-normalized amplitude balance.

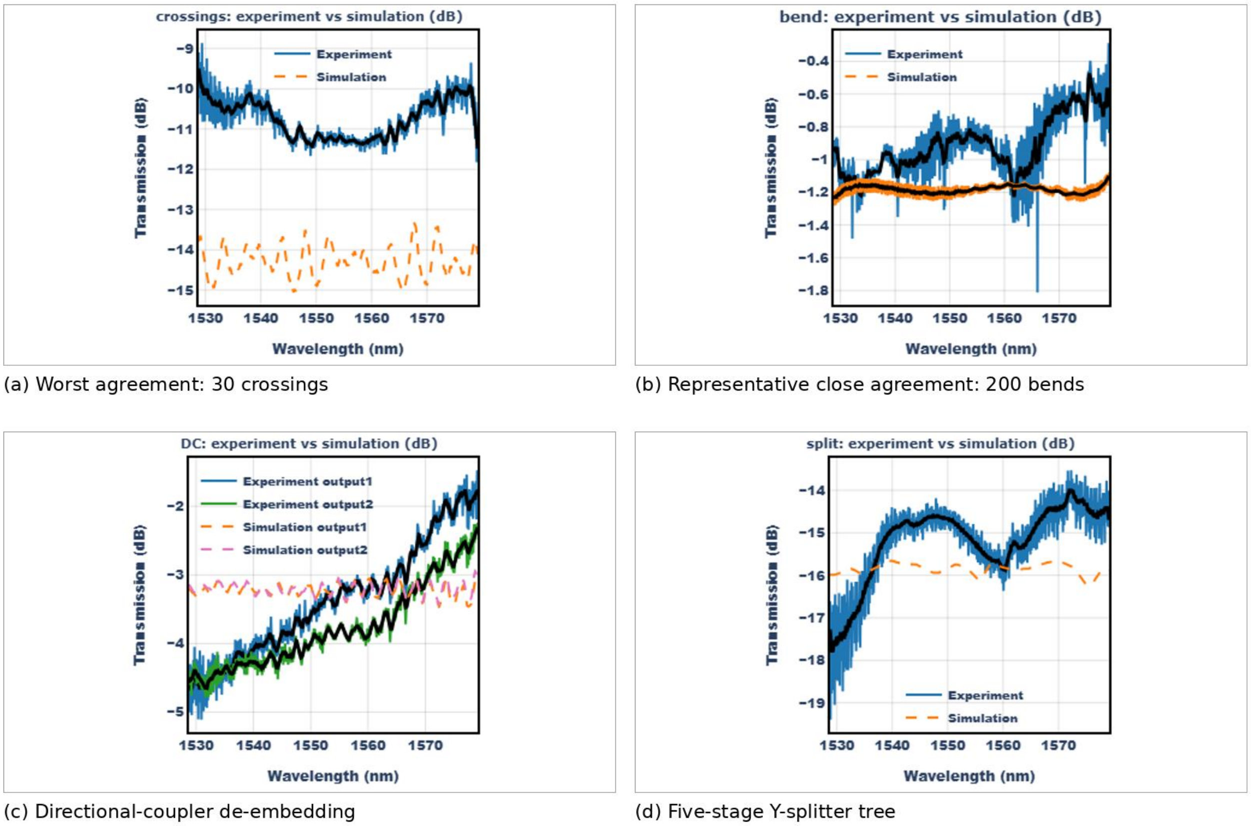
4.2 Ring and reference structures

The ring resonator and the reference structure broaden the evidence base around the MZI result. The reference spectrum provides the broadband grating-coupler envelope needed for normalization, while the ring spectrum introduces a train of resonant notches that depends on the same underlying waveguide group index. Using the corrected nominal waveguide baseline and a 10 um ring radius gives an expected ring FSR of approximately 9.10 nm at 1550 nm. The measured notch spacing is visually consistent with that scale, so the ring behaves as a resonant corroboration of the same platform dispersion that the MZI probes interferometrically.

4.3 Passive-component benchmark and loss extraction

The passive benchmark structures isolate the main compact-model building blocks used elsewhere on the chip. Bends, crossings, splitters, and the directional coupler were normalized with the reference envelope and compared directly against their simulation counterparts. MAE is reported in dB, and next to it the report gives the mean linear-power deviation in percent so that each error number has an immediately interpretable power-domain counterpart.

Representative experiment-versus-simulation overlays



(a) Worst agreement: 30 crossings

(b) Representative close agreement: 200 bends

(c) Directional-coupler de-embedding

(d) Five-stage Y-splitter tree

Figure 5. Representative experiment-versus-simulation overlays. The 30-crossing structure is the worst agreement case, while the 200-bend structure is a representative close match. The directional coupler and the five-stage Y-splitter tree illustrate the intermediate regime.

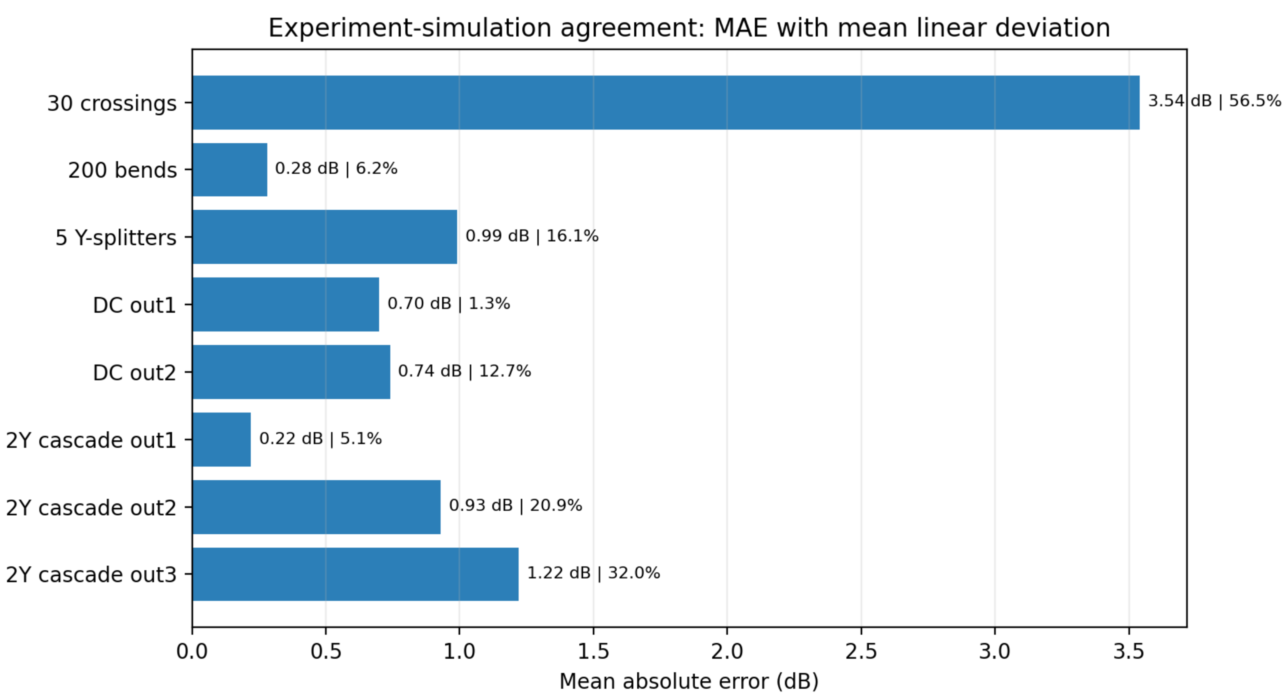


Figure 6. Experiment-simulation agreement summary. Each bar reports MAE in dB together with the corresponding mean linear-power deviation in percent.

Structure	Exp. mean [dB]	Sim. mean [dB]	MAE [dB]	Deviation [%]
30 crossings	-10.70	-14.30	3.54	56.5
200 bends	-0.92	-1.19	0.28	6.2
5 Y-splitters	-15.20	-15.80	0.99	16.1
Directional coupler output 1	-3.30	-3.20	0.70	1.3
Directional coupler output 2	-3.70	-3.20	0.74	12.7
2Y cascade output 1	-3.67	-3.45	0.22	5.1
2Y cascade output 2	-7.27	-6.45	0.93	20.9
2Y cascade output 3	-7.57	-6.36	1.22	32.0

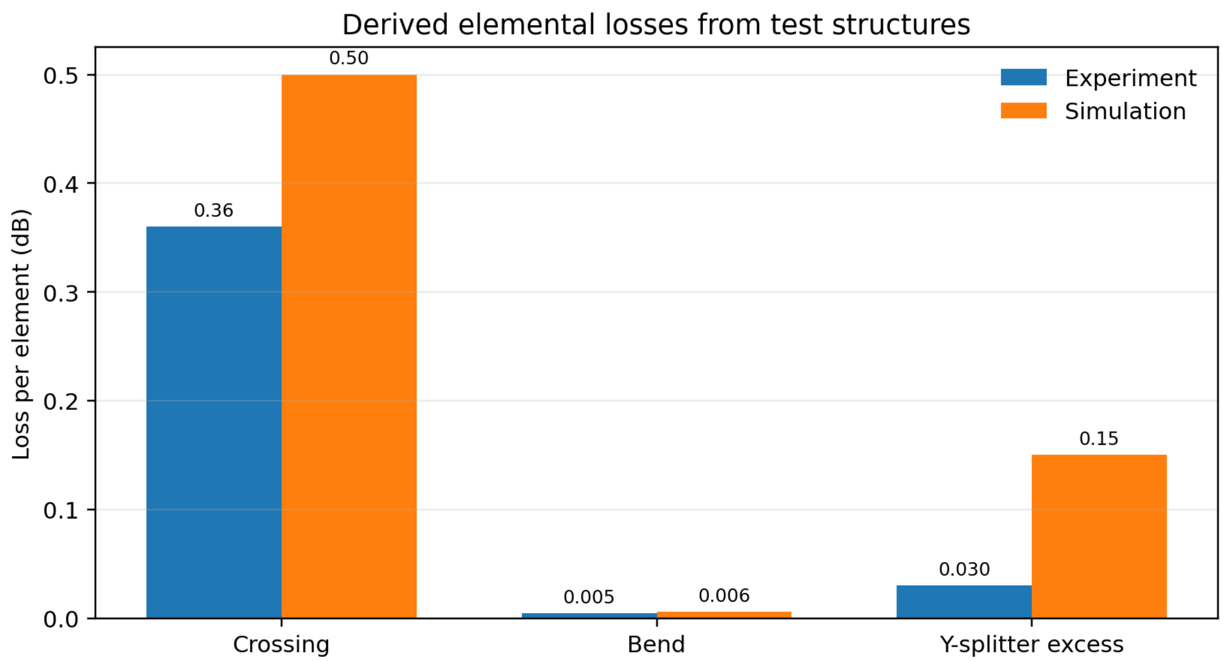


Figure 7. Elemental losses derived from the dedicated benchmark structures. Crossings dominate the total mismatch; bends are already close to simulation.

Element	Experiment	Simulation
Crossing loss [dB/element]	0.36	0.50
Bend loss [dB/element]	0.00455	0.00589
Y-splitter excess [dB/element]	0.03	0.15

The ranking is unambiguous. The best match is 2Y cascade output 1 at 0.22 dB MAE, closely followed by the 200-bend structure at 0.28 dB MAE. The five-stage Y-splitter tree and the directional-coupler outputs occupy the middle of the range. The worst case is the 30-crossing structure, which is separated from the rest of the set by a large margin. Terminators are not modelled, thus giving a more pessimistic transmission spectrum than the experiment.

4.4 Additional structures: phase-shift MZI, MMI 1x4, and 2Y cascade

The pi-phase-shift MZI: the measured null at 1550 nm is around -16.2 dB, whereas the simulated null is far deeper. That result shows how quickly small phase or coupler-balance errors become large transmission errors in a phase-targeted destructive-interference design.

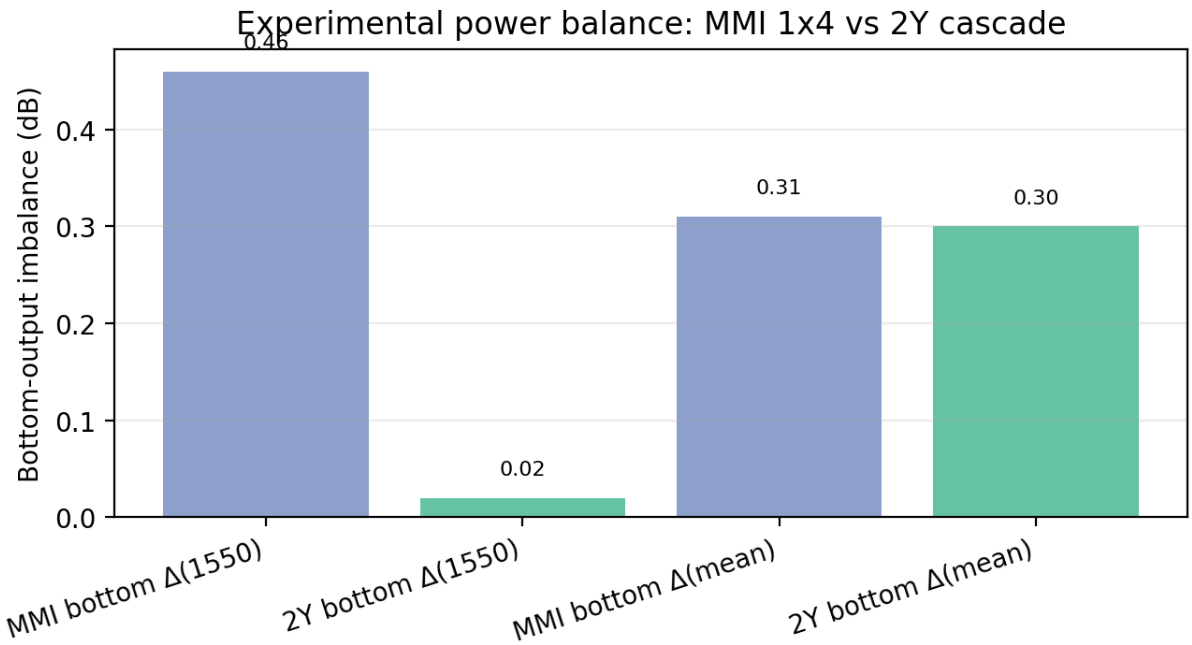


Figure 8. Experimental bottom-output balance comparison between the MMI 1x4 and the two-stage Y-splitter cascade.

The MMI 1x4 nevertheless demonstrates the intended qualitative behavior: the top port is strongly suppressed, and the bottom branches land near the expected -6 to -7 dB level. The two-stage Y-splitter cascade gives a smaller bottom-output imbalance at 1550 nm, but its top-port behavior is intentionally constructive rather than destructive, so the two devices are not interchangeable. Together they show that output balance can be engineered successfully, but the most phase-sensitive top-port predictions remain more fragile than broadband loss quantities such as bend loss.

5. Conclusion

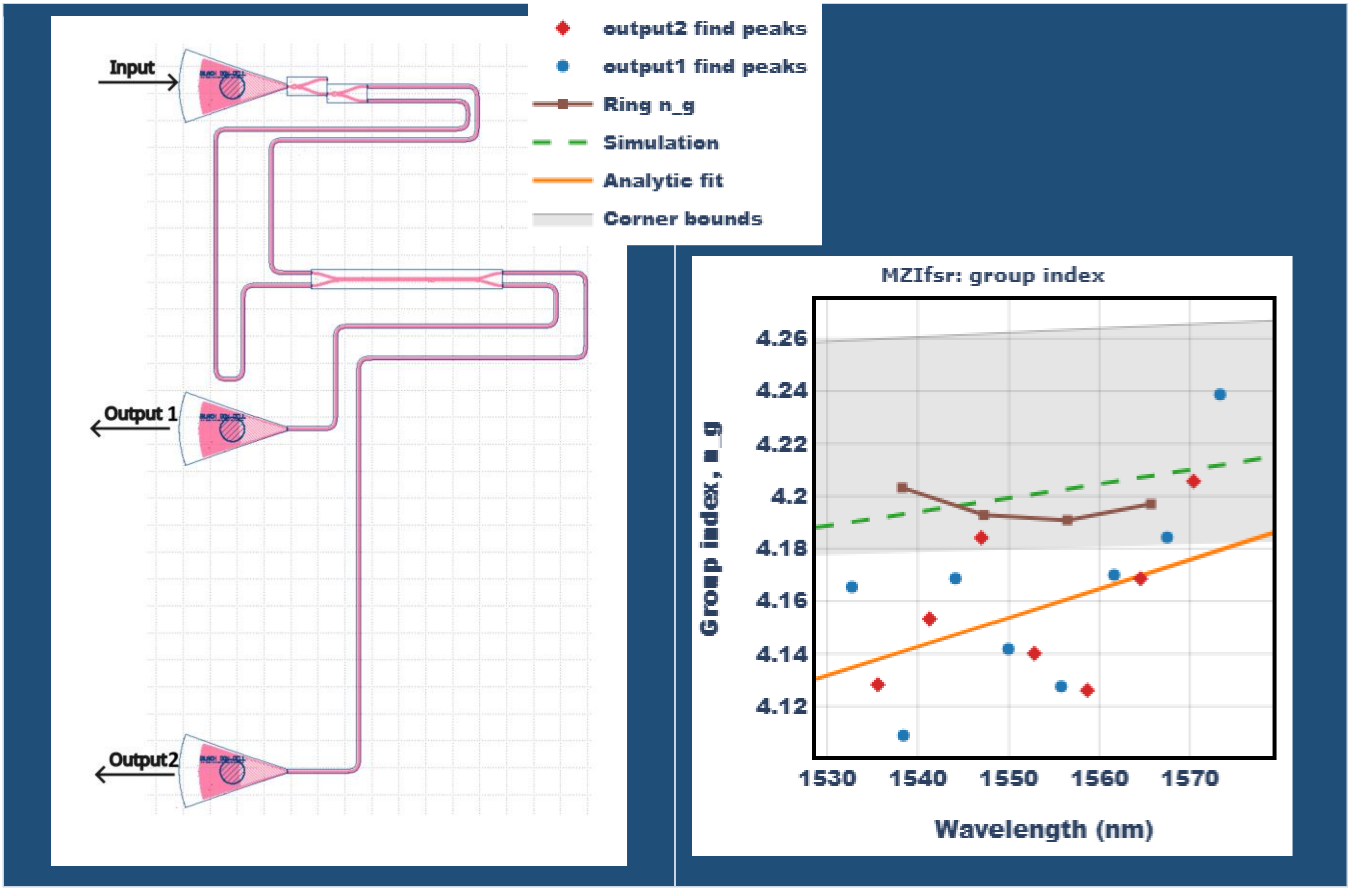
The fabricated SOI chip successfully supports both main objectives of the study. The unbalanced MZI with Delta L = 100 um provides the primary group-index extraction route, yielding a paired-port analytic value of n_g(1550 nm) = 4.1536 with complementary minima-based checkpoints of 4.1417 and 4.1615. The corrected nominal waveguide simulation baseline from the uploaded text files gives n_g(1550 nm) = 4.2019 and a nominal MZI FSR of 5.72 nm, while the ring resonator provides an additional resonance-based consistency check with an expected FSR of approximately 9.10 nm.

The passive benchmark also succeeds in translating spectra into component-level quantities. The extracted losses are 0.36 dB per crossing in experiment versus 0.50 dB in simulation, 0.00455 versus 0.00589 dB per bend, and 0.03 versus 0.15 dB excess loss per Y-splitter. Several structures already match closely, especially the 200-bend device and the best two-stage Y-splitter path, while the main remaining gap is concentrated in the crossing model and in highly phase-sensitive interference targets.

Appendix A. Structure-by-structure layout and analysis gallery

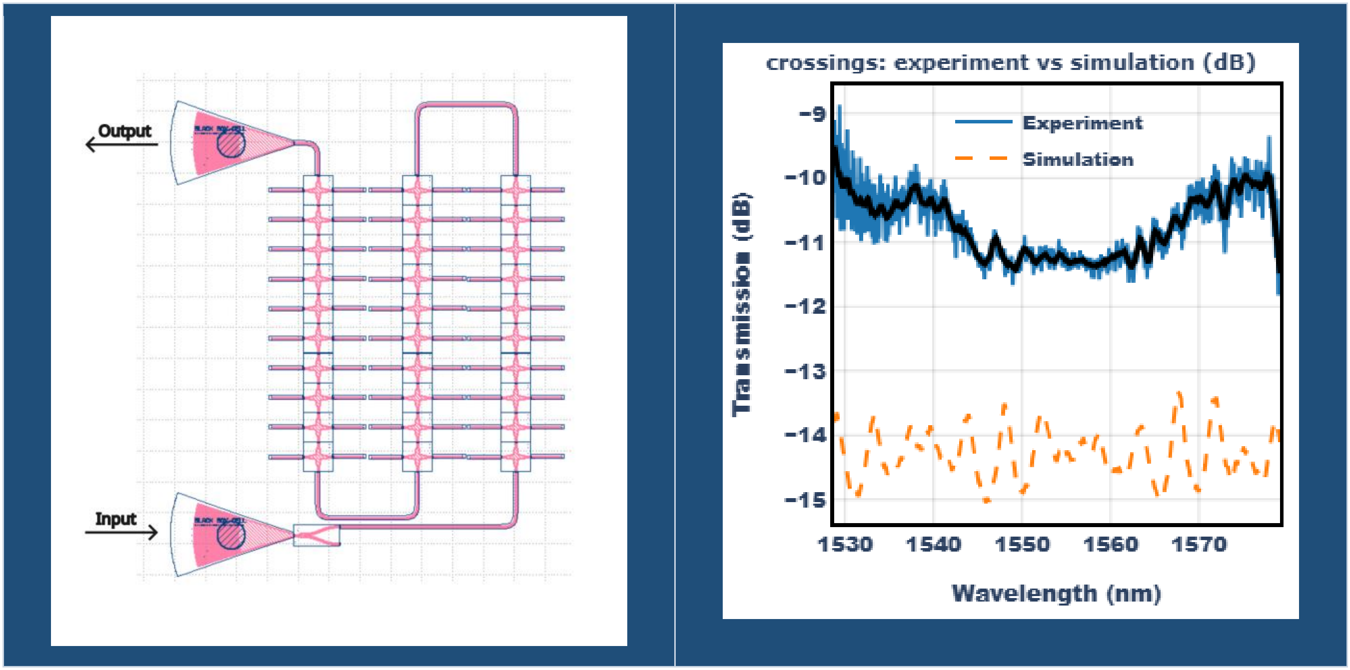
Each subsection below pairs a layout image with a representative analysis plot.

Unbalanced MZI (Delta L = 100 um)



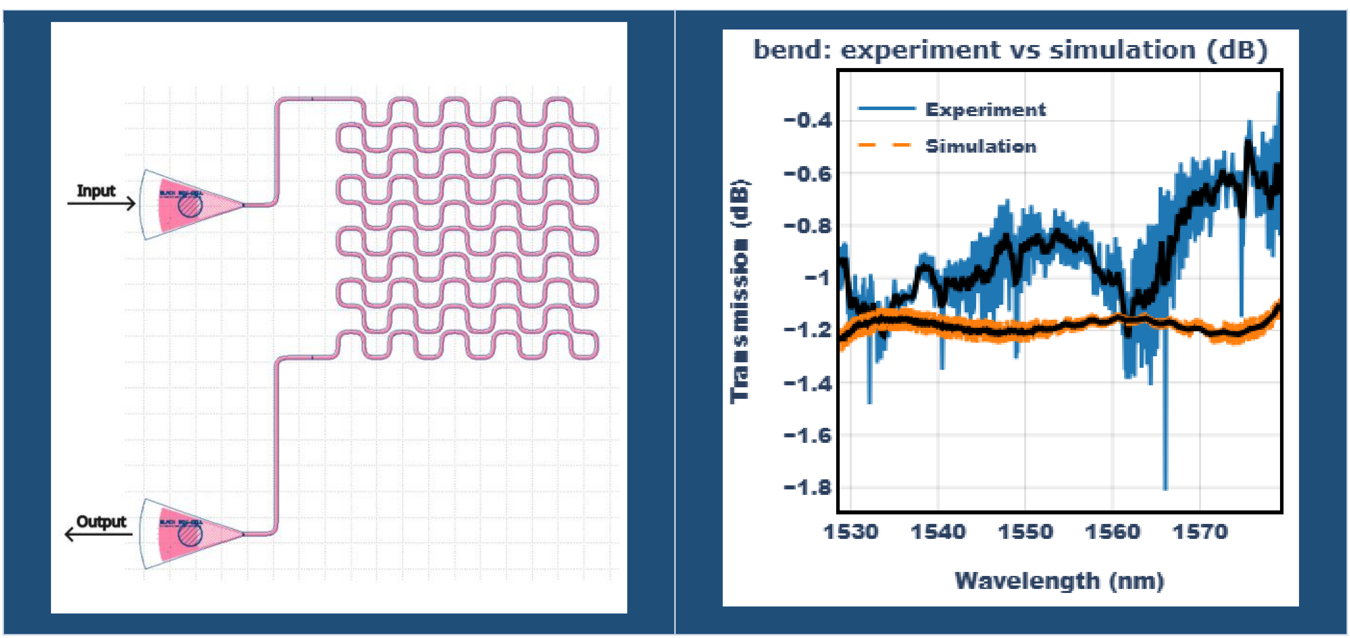
Purpose: extract group index from FSR. The paired-port analytic fit gives n_g(1550 nm) = 4.1536, while the two minima-based estimates are 4.1417 and 4.1615.

30 crossings



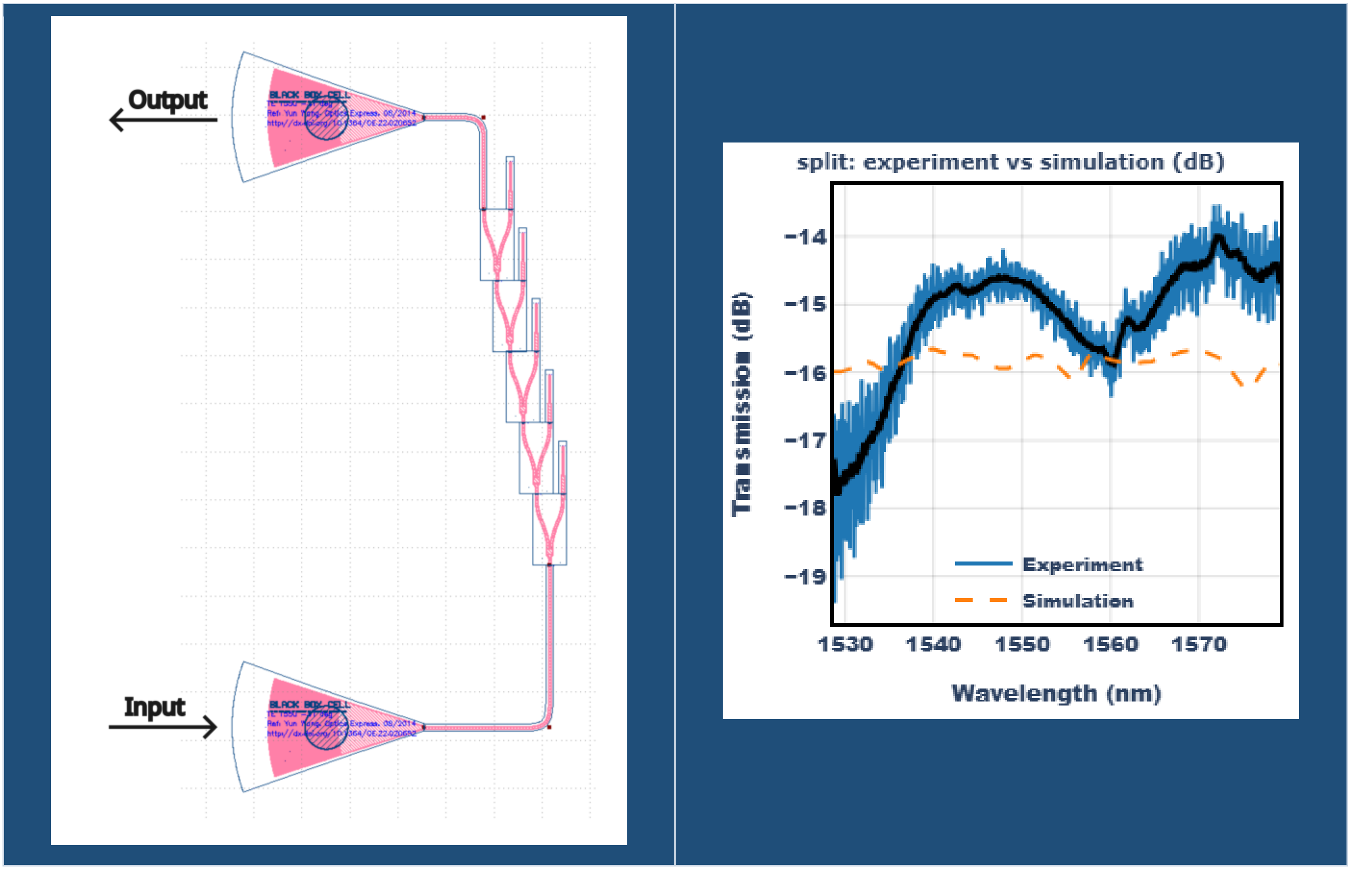
Purpose: crossing-loss extraction. This is the largest experiment-simulation mismatch in the set, with 3.54 dB MAE and approximately 56.5% mean linear-power deviation.

200 bends



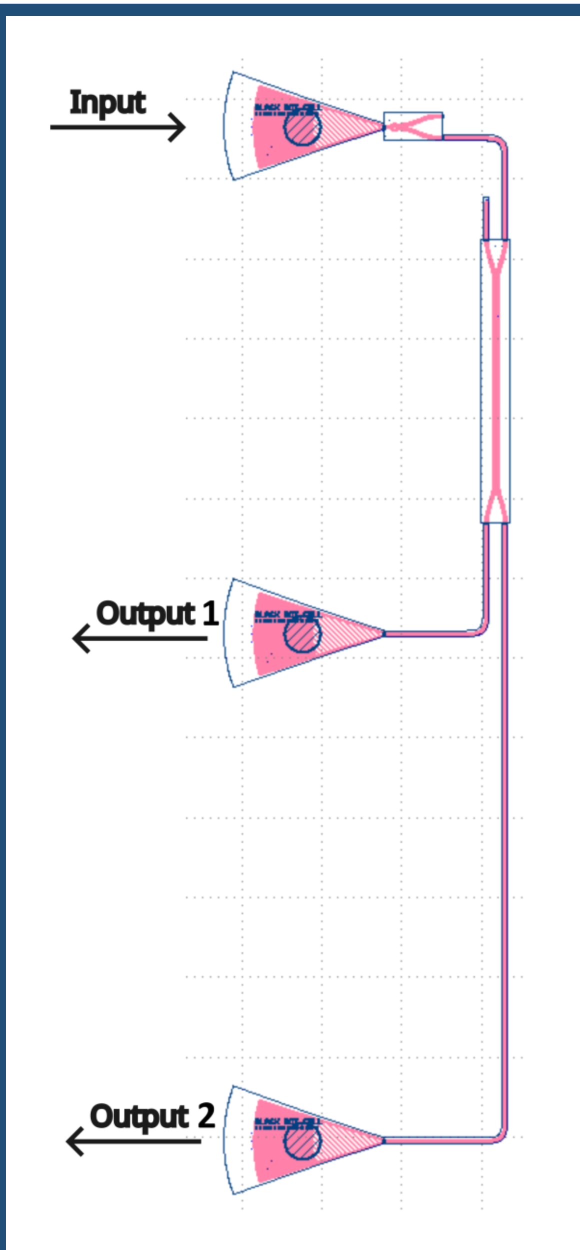
Purpose: bend-loss extraction. Agreement is close, with 0.28 dB MAE and approximately 6.2% mean linear-power deviation.

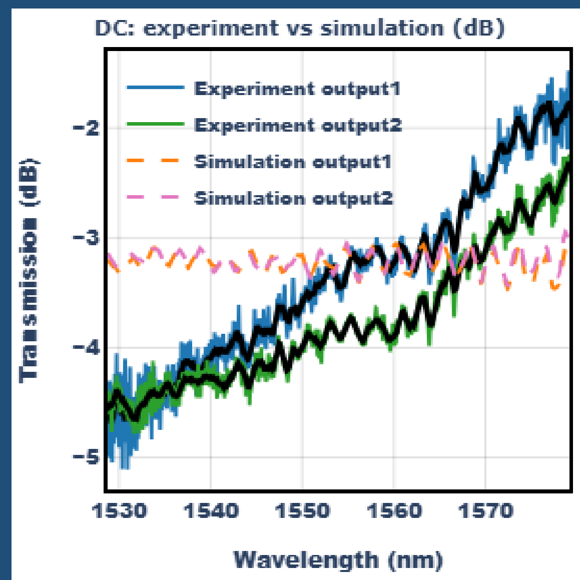
Five-stage Y-splitter tree



Purpose: Y-splitter excess-loss extraction. The measured mean transmission is -15.2 dB versus -15.8 dB in simulation, corresponding to 0.99 dB MAE.

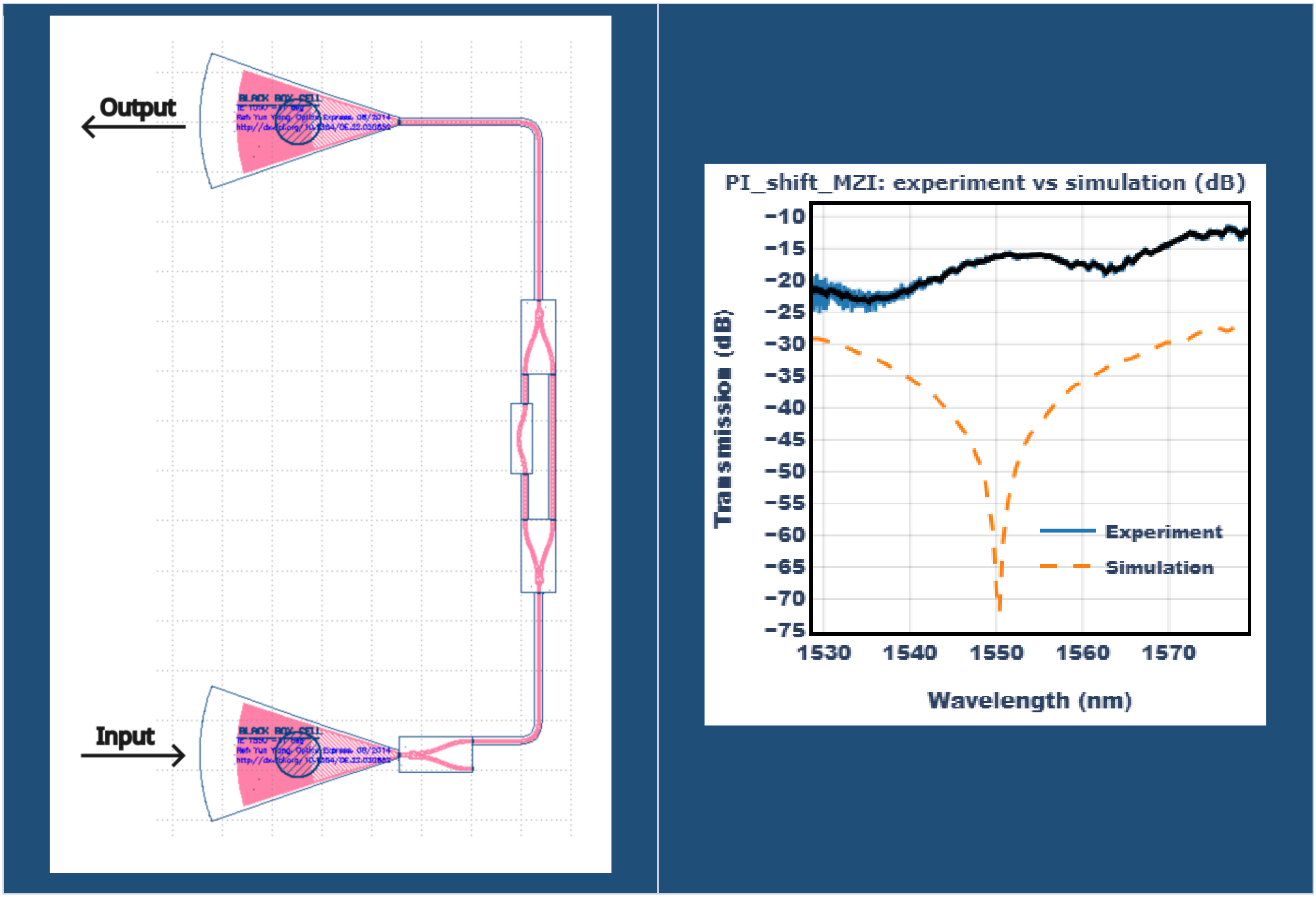
Adiabatic directional coupler





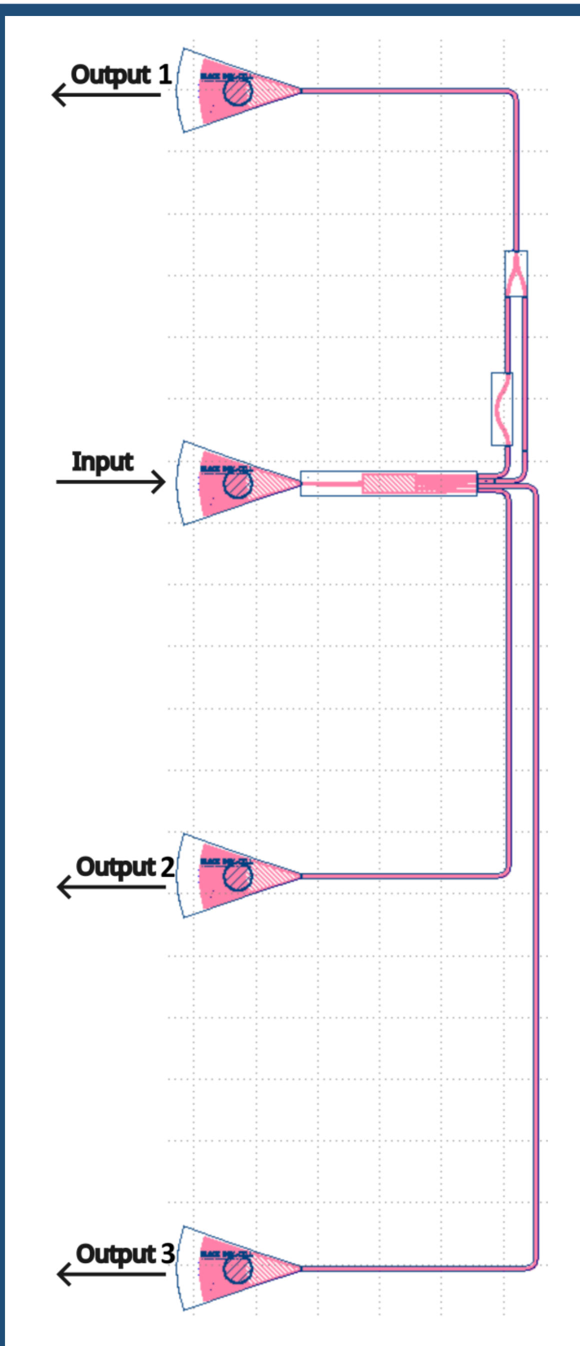
Purpose: de-embed the coupler spectrum used in the unbalanced MZI. Output 1 and output 2 show 0.70 dB and 0.74 dB MAE, respectively.

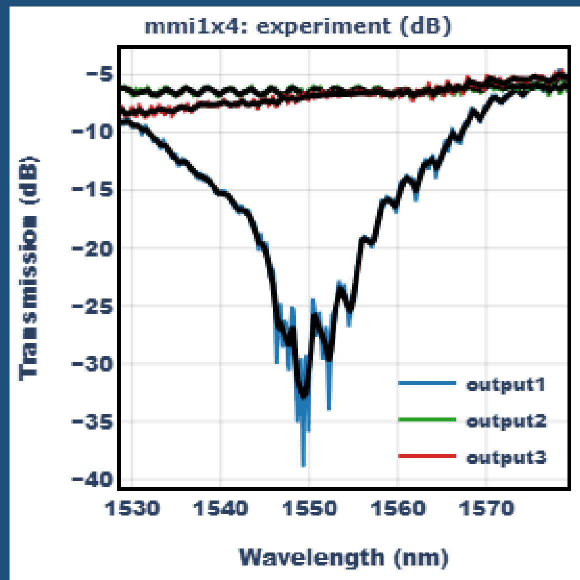
Pi-phase-shift MZI



Purpose: test phase-targeted interference. The measured null near 1550 nm is much shallower than the simulated one, highlighting the sensitivity of phase-engineered nulls to fabrication and model error.

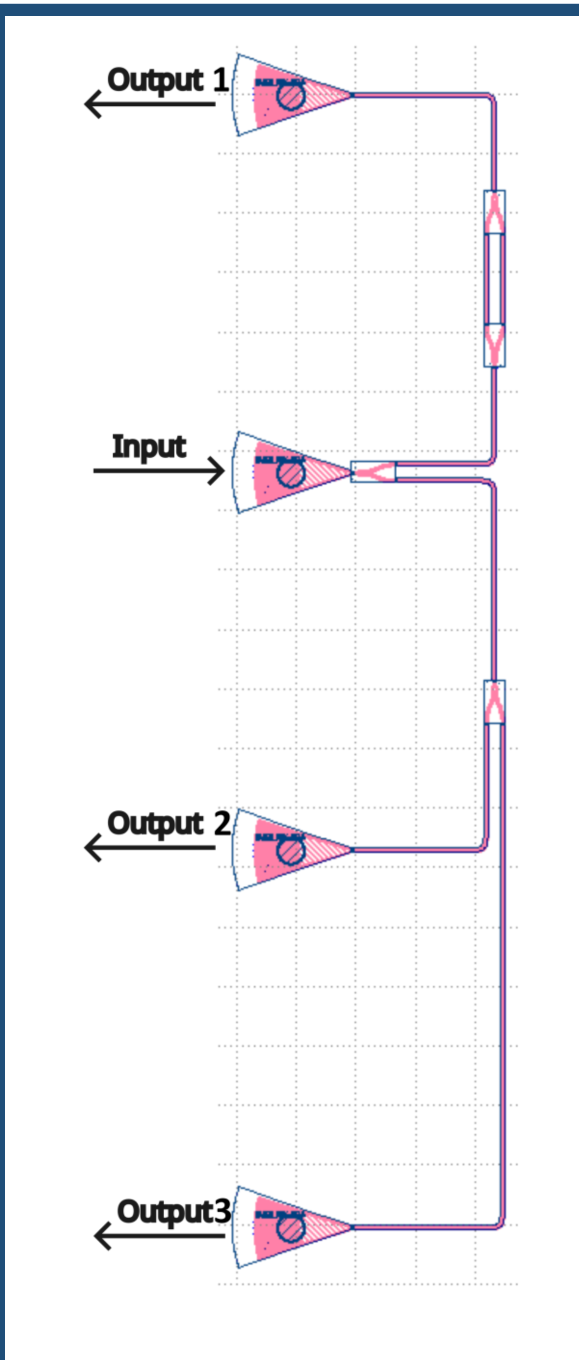
MMI 1x4

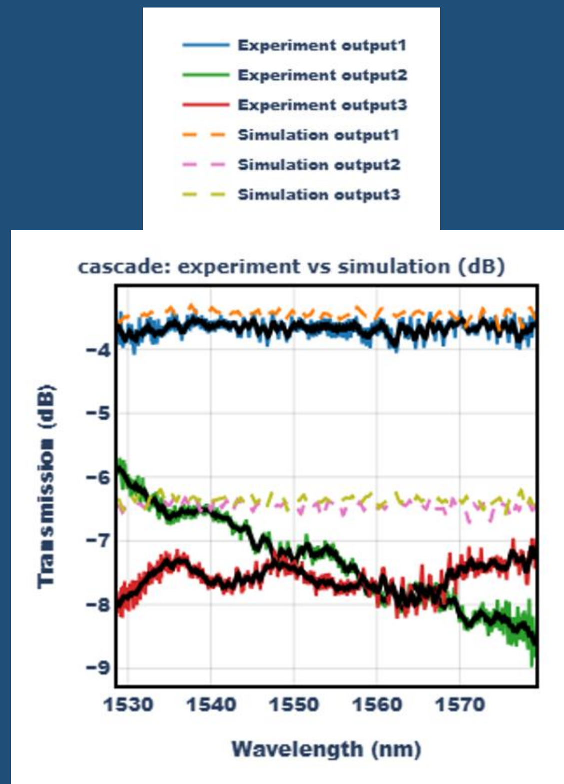




Purpose: evaluate output balance and top-port suppression. The top port is strongly suppressed, while the two bottom outputs differ by 0.46 dB at 1550 nm.

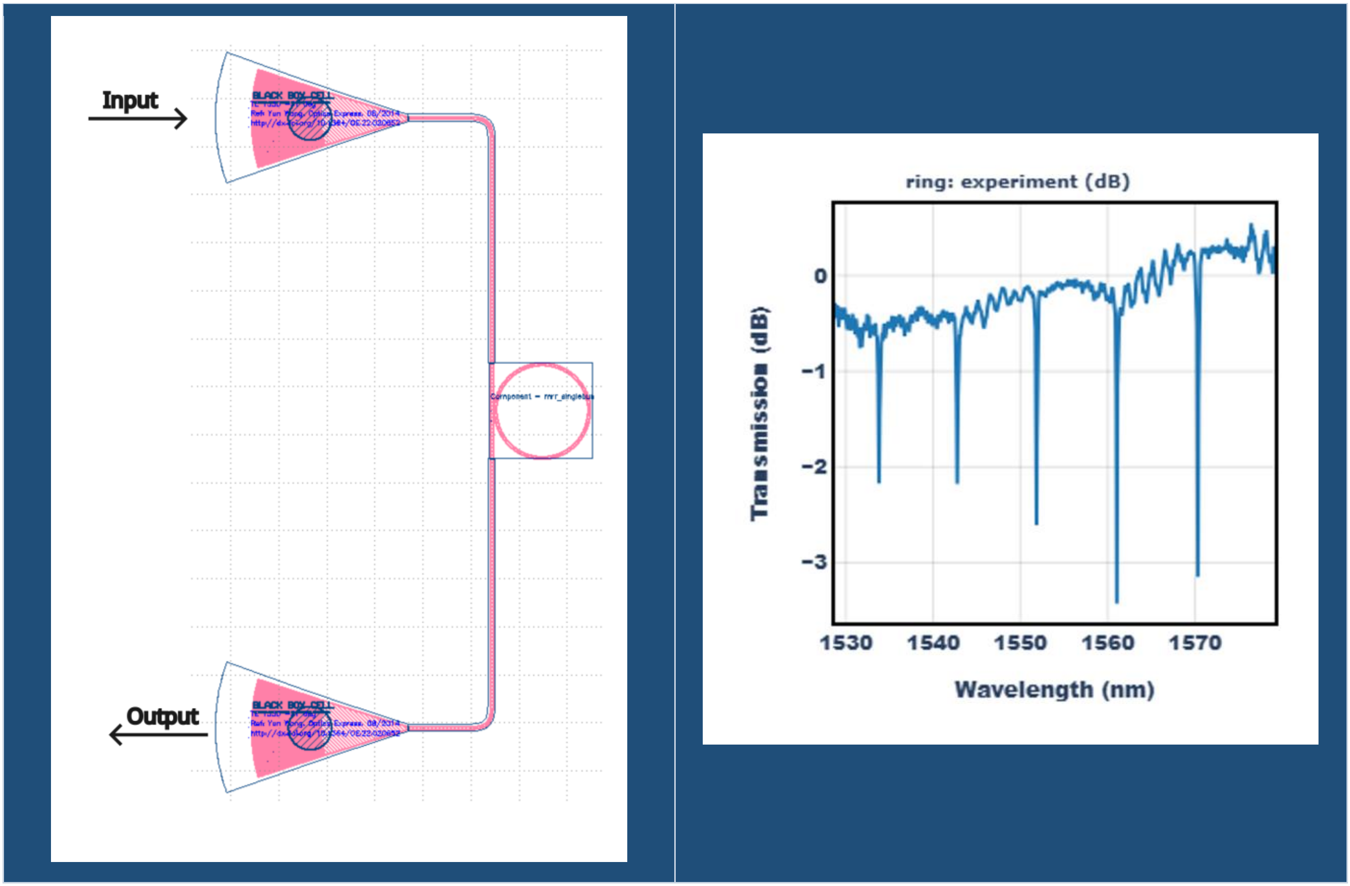
Two-stage Y-splitter cascade





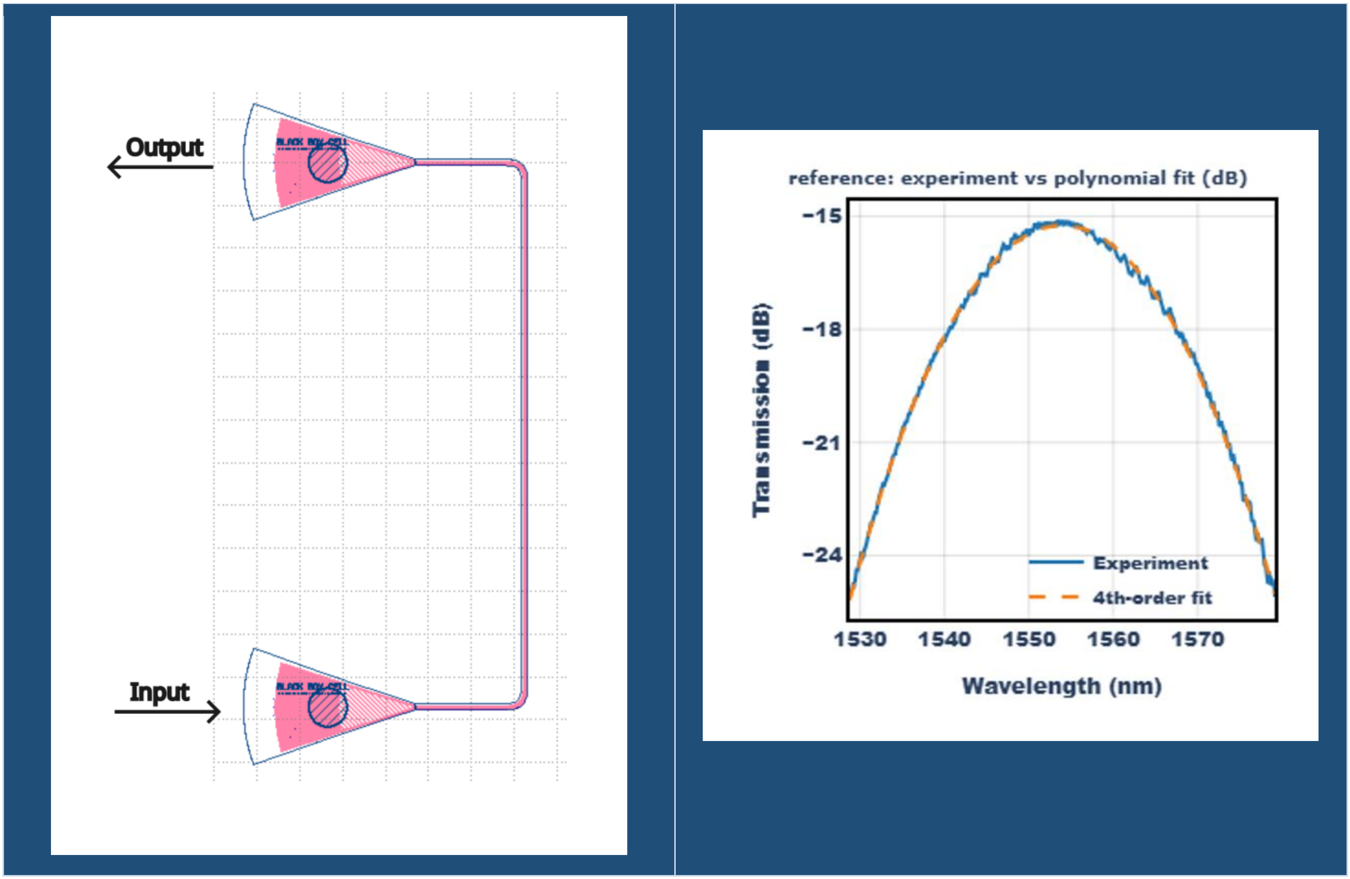
Purpose: compare the MMI 1x4 against a Y-splitter-based alternative. The top output is constructive near -3.7 dB, and the bottom two outputs are closely matched.

Single-bus ring resonator



Purpose: independent resonant route to group-index extraction. The resonance spacing is visually consistent with the approximately 9.10 nm FSR predicted from the updated nominal waveguide model.

Reference structure



Purpose: capture the grating-coupler transmission envelope. This spectrum is fit and subtracted in dB so that device-level responses can be compared on a normalized basis.

References

Chrostowski, L., and Hochberg, M., Silicon Photonics Design: From Devices to Systems, Cambridge University Press, 2015.

Capmany, J., and Perez, D., Programmable Integrated Photonics, Oxford University Press, 2020.